

InEar Teensy Compact Protocol

Data Protocol Specification

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1 Protocol Overview

The Compact Protocol is a lean, variable-length binary streaming format for real-time EEG acquisition from an ADS1299-based biosignal amplifier on a Teensy 4.1 microcontroller. It improves on earlier InEar protocols by removing every non-essential byte while adding built-in TTL support and error signaling.

Key design decisions:

- **No footer bytes** — the 2-byte 0xC0 0xC0 end marker is eliminated entirely. The parser relies on the sync header + known packet size.
- **No sequence counter** — the 4-byte microsecond timestamp provides both ordering and loss detection.
- **TTL merged into TYPE byte** — two TTL bits (input + output) share a byte with the 3-bit packet type, saving the former 1-byte marker field.
- **EEG + at most one auxiliary sensor per packet** — staggered scheduling avoids large combined packets.
- **Built-in error signaling** — a reserved bit pattern forces the host application back into handshake mode.

Table 1: Protocol parameters at a glance

Parameter	Value
EEG Sample Rate	1000 Hz
EEG Channels	8 (24-bit signed, 24 B total)
Baud Rate	2,000,000 (2 Mbaud)
Byte Order	Big-Endian (network order)
Packet Size	32–38 B (variable, depends on aux payload)
Best Case (EEG only)	32 B
Worst Case (EEG + Accel)	38 B
Fixed Overhead	8 B (7 B header + 1 B checksum)
Error Detection	XOR checksum (1 B)

2 Packet Structure

Every packet follows the same skeleton:

START (2 B)		TIMESTAMP (4 B)		TYPE+TTL (1 B)		EEG (24 B)		[AUX]		CHECKSUM (1 B)
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There is **no footer**. The packet ends immediately after the checksum byte.

2.1 Header (7 bytes — always present)

Table 2: Header fields

Offset	Size	Field	Format	Description
0	1 B	Sync 1	0xA5	Header byte 1 (constant)
1	1 B	Sync 2	0x5A	Header byte 2 (constant)
2–5	4 B	Timestamp	uint32 BE	Microseconds since boot
6	1 B	Type+TTL	bitfield	Packet type + TTL lines (section 5)
Total:		7 bytes		

2.2 EEG Data (24 bytes — always present)

Eight channels of 24-bit signed big-endian data from the ADS1299:

Table 3: EEG data block (8 channels $\times$ 3 B = 24 B)

Offset	Size	Field	Format	Description
7–9	3 B	EEG Ch 1	int24 BE	Channel 1
10–12	3 B	EEG Ch 2	int24 BE	Channel 2
13–15	3 B	EEG Ch 3	int24 BE	Channel 3
16–18	3 B	EEG Ch 4	int24 BE	Channel 4
19–21	3 B	EEG Ch 5	int24 BE	Channel 5
22–24	3 B	EEG Ch 6	int24 BE	Channel 6
25–27	3 B	EEG Ch 7	int24 BE	Channel 7
28–30	3 B	EEG Ch 8	int24 BE	Channel 8
Total: 24 bytes (offsets 7–30)				

2.3 Optional Auxiliary Block (0–6 bytes)

At most **one** auxiliary sensor is attached per packet. The aux block immediately follows the EEG data at offset 31. The TYPE field (bits 2–0) determines which block, if any, is present:

Table 4: Auxiliary data blocks by type code

Type	Aux Block	Size	Format	Sensor IC
0x00	(none)	0 B	—	—
0x01	Accelerometer	6 B	3 $\times$ int16 BE	ADXL367
0x02	PPG + SpO ₂	4 B	uint24 BE + uint8	MAXM86161
0x03	Temperature	2 B	int16 BE	TMP117
0x04	Battery	1 B	uint8	(voltage divider)
0x05	Reserved 1	—	—	—
0x06	Reserved 2	—	—	—
0x07	Reserved 3	—	—	—

2.4 Checksum (1 byte — always last)

XOR of every preceding byte in the packet (offset 0 to $n-2$, where n is the total packet length):

Offset	Size	Field	Format	Description
$n-1$	1 B	Checksum	uint8	XOR of bytes 0.. $(n-2)$

3 Packet Size Summary

Table 5: Packet sizes for each type code

Type	Contents	Header	Aux	Total
0x00	EEG only (+ TTL)	7 + 24 + 1	0 B	32 B
0x01	EEG + Accelerometer	7 + 24 + 1	6 B	38 B
0x02	EEG + PPG + SpO ₂	7 + 24 + 1	4 B	36 B
0x03	EEG + Temperature	7 + 24 + 1	2 B	34 B
0x04	EEG + Battery	7 + 24 + 1	1 B	33 B
0x05	Reserved 1	—	—	—
0x06	Reserved 2	—	—	—
0x07	Reserved 3	—	—	—

4 Packet Structure Diagrams

The diagrams below show the field layout of each packet type. Field widths are **schematic** (not proportional to byte count); see section 6 for exact byte offsets.

4.1 Best Case — EEG + TTL Only (32 B)

START A5 5A 2 B	TSMP uint32 BE 4 B	TYPE +TTL 1 B	EEG — 8 channels $\times$ 3 B = 24 B	CHK XOR 1 B
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Figure 1: EEG-only packet (Type 0x00) — 32 bytes. Most common packet (~80% of all packets).

4.2 Worst Case — EEG + Accelerometer (38 B)

START A5 5A 2 B	TSMP uint32 BE 4 B	TYPE +TTL 1 B	EEG — 8 ch $\times$ 3 B = 24 B	ACCEL X, Y, Z 6 B	CHK XOR 1 B
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Figure 2: EEG + Accelerometer packet (Type 0x01) — 38 bytes. Largest packet in the protocol.

4.3 EEG + PPG + SpO₂ (36 B)

START A5 5A 2 B	TSMP uint32 BE 4 B	TYPE +TTL 1 B	EEG — 8 ch $\times$ 3 B = 24 B	PPG uint24 3 B	SpO₂ uint8 1 B	CHK XOR 1 B
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Figure 3: EEG + PPG + SpO₂ packet (Type 0x02) — 36 bytes.

4.4 EEG + Temperature (34 B)

START A5 5A 2 B	TSMP uint32 BE 4 B	TYPE +TTL 1 B	EEG — 8 ch $\times$ 3 B = 24 B	TEMP int16 BE 2 B	CHK XOR 1 B
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Figure 4: EEG + Temperature packet (Type 0x03) — 34 bytes.

4.5 EEG + Battery (33 B)

START A5 5A 2 B	TSMP uint32 BE 4 B	TYPE +TTL 1 B	EEG — 8 ch $\times$ 3 B = 24 B	BAT uint8 1 B	CHK XOR 1 B
------------------------------	---------------------------------	-----------------------------------	---------------------------------------	----------------------------	--------------------------

Figure 5: EEG + Battery packet (Type 0x04) — 33 bytes.

5 TYPE+TTL Byte

The single TYPE+TTL byte at offset 6 packs both TTL trigger lines and the packet type into 8 bits:

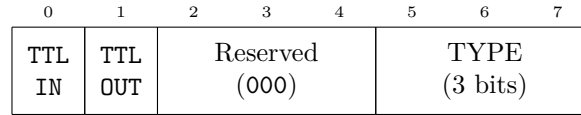


Figure 6: TYPE+TTL byte bit-field layout. Bit 7 = MSB, Bit 0 = LSB.

5.1 Bit Assignments

Table 6: TYPE+TTL bit assignments

Bit(s)	Name	Description
7	TTL_IN	External trigger input (1 = active)
6	TTL_OUT	External trigger output (1 = active)
5–3	Reserved	Must be 000 for normal packets; 111 = error signal
2–0	TYPE	Packet type selector (0–7)

5.2 Type Code Chart

Table 7: Type codes (bits 2–0 of TYPE+TTL byte)

Binary	Hex	Name	Auxiliary Data	Packet Size
000	0x00	EEG	None	32 B
001	0x01	EEG_ACCEL	Accelerometer (6 B)	38 B
010	0x02	EEG_PPG	PPG + SpO ₂ (4 B)	36 B
011	0x03	EEG_TEMP	Temperature (2 B)	34 B
100	0x04	EEG_BAT	Battery level (1 B)	33 B
101	0x05	RESERVED_1	Future use	—
110	0x06	RESERVED_2	Future use	—
111	0x07	RESERVED_3	Future use	—

5.3 TTL Examples

Table 8: Example TYPE+TTL byte values

Hex	Binary	TTL	Type	Meaning
0x00	0 0 000 000	—	EEG	EEG only, no trigger
0x80	1 0 000 000	IN	EEG	EEG only, TTL input active
0xC1	1 1 000 001	IN+OUT	ACCEL	EEG + Accel, both triggers
0x02	0 0 000 010	—	PPG	EEG + PPG + SpO ₂ , no trigger
0x43	0 1 000 011	OUT	TEMP	EEG + Temp, TTL output active
0x84	1 0 000 100	IN	BAT	EEG + Battery, TTL input active

5.4 Error Signaling

When the reserved bits (5–3) are all set to 1, the packet signals an **error condition**:

$$\text{Error byte} = ?? \underbrace{111}_{\text{error flag}} ???_{\text{binary}} \longrightarrow \text{bits 5–3} = 111$$

An error packet consists of the normal header structure with the error flag set and zeroed-out EEG data:

A5 5A + TIMESTAMP (4 B) + Error-flagged TYPE+TTL + 24×0x00 + CHECKSUM

Upon receiving a packet with bits 5–3 = 111, the Qt host application must:

1. Stop data acquisition immediately.
2. Transition back to the handshake / connection state.
3. Report the error condition to the user.

6 Byte Offset Reference

Table 9: Absolute byte offsets for all active packet types (8-channel EEG)

Block	0x00 EEG	0x01 +Accel	0x02 +PPG	0x03 +Temp	0x04 +Bat
Sync (2 B)	0–1	0–1	0–1	0–1	0–1
Timestamp (4 B)	2–5	2–5	2–5	2–5	2–5
Type+TTL (1 B)	6	6	6	6	6
EEG (24 B)	7–30	7–30	7–30	7–30	7–30
Aux data	—	31–36	31–34	31–32	31
Checksum (1 B)	31	37	35	33	32
Total	32 B	38 B	36 B	34 B	33 B

7 Multi-Rate Sensor Scheduling

Every packet contains **fresh** EEG data sampled at 1000 Hz. Each auxiliary sensor has its own dedicated packet type and sub-rate. Because the protocol allows at most one auxiliary block per packet, collisions between sensors due at the same sample are resolved by **deferring** the lower-priority sensor by 1 ms. The deferred packet still carries a fresh EEG sample from that later time slot — no EEG data is ever stale or duplicated.

Table 10: Sensor scheduling rates

Sensor	Rate (Hz)	Every <i>N</i> th pkt	Justification
EEG (+ TTL)	1000	Every packet	Fast-changing neural signals
Accelerometer	200	Every 5th	Head movement, artifact rejection
PPG + SpO ₂	100	Every 10th	Cardiac rhythm (~25 Hz BW)
Temperature	1	Every 1000th	Near-DC body temperature
Battery	1	Every 1000th	Slowly changing level

7.1 Collision Resolution: 1 ms Deferral

When two or more auxiliary sensors are due on the same sample, the highest-priority sensor claims that packet and lower-priority sensors are deferred by 1 ms (one sample period). The deferred packet is a normal packet of the appropriate type, carrying its own fresh EEG sample from the deferred time slot.

Priority order (highest first):

$$\text{ACCEL} > \text{PPG} > \text{TEMP} > \text{BAT}$$

This guarantees that every packet carries at most one auxiliary block and that **no EEG sample is ever duplicated or skipped** — each deferred packet simply uses the next 1 ms EEG reading.

Listing 1: Deferral timing: PPG deferred by 1 ms when Accel has priority

Time (ms):	...	9	10	11	12	...
Accel due?		no	YES	no	no	
PPG due?		no	YES	no	no	
Result:	...	[EEG]	[ACCEL]	[PPG]	[EEG]	...
			^fresh	^fresh	EEG from t=11 ms	
			EEG			
			t=10			

7.2 1 Hz Sensor Staggering

Temperature and Battery are both sampled at 1 Hz (every 1000th packet). To avoid them always colliding, they are staggered with a fixed offset so that only one 1 Hz sensor is due at any given second boundary.

Note: The exact staggering offset between Temperature and Battery is **TBD** (not yet finalized in the firmware). The only hard constraint is:

- Temperature and Battery must not be due on the same sample.
- If either 1 Hz sensor collides with an Accel or PPG packet, the 1 Hz sensor is deferred by 1 ms (same deferral rule as above).

7.3 Example: 21 Consecutive Packets

Listing 2: Scheduling over 21 samples (Accel every 5th, PPG every 10th)

Pkt#:	1	2	3	4	5	6	7	8	9	10	
EEG:	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	
Aux:	---	---	---	---	ACCEL	---	---	---	---	ACCEL	
Size:	32B	32B	32B	32B	38B	32B	32B	32B	32B	38B	
Pkt#:	11	12	13	14	15	16	17	18	19	20	21
EEG:	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]	[EEG]
Aux:	PPG	---	---	---	ACCEL	---	---	---	---	ACCEL	PPG
Size:	36B	32B	32B	32B	38B	32B	32B	32B	32B	38B	36B
Note:	PPG due at pkt 10 deferred to pkt 11 (Accel had priority).										
	PPG due at pkt 20 deferred to pkt 21 (Accel had priority).										
	Each deferred packet carries fresh EEG from its own time slot.										

7.4 Bandwidth Estimate

Assuming 200 Hz accelerometer, 100 Hz PPG, and 1 Hz each for temperature and battery:

$$\begin{aligned}
 \text{Approx. avg B/s} &\approx 698 \times 32 + 200 \times 38 + 100 \times 36 + 1 \times 34 + 1 \times 33 \\
 &= 22,336 + 7,600 + 3,600 + 34 + 33 \\
 &\approx \mathbf{33,603 \text{ B/s}}
 \end{aligned}$$

$$\text{Serial utilization: } \frac{33,603}{200,000} = \mathbf{16.8\%} \quad (\text{headroom: } 83.2\%)$$

7.5 Overhead Efficiency by Channel Count

Table 11: Fixed 8-byte overhead becomes proportionally cheaper with more EEG channels

EEG Channels	EEG Size	Best-Case Packet	Overhead %
5	15 B	23 B	34.8%
8	24 B	32 B	25.0%
12	36 B	44 B	18.2%
16	48 B	56 B	14.3%

8 LSL Stream Metadata Reference

Lab Streaming Layer (LSL) is used to transport data from Open Ephys to downstream analysis tools (LabRecorder, MNE-Python, EEGLAB). Correct metadata is critical for interoperability and BIDS-compliant recordings.

8.1 StreamInfo Properties

Table 12: LSL StreamInfo constructor parameters

Property	EEG Stream	Marker Stream	Notes
name	"InEarEEG"	"InEarEEG_Markers"	Human-readable name. Must be unique on the network if multiple devices are present.
type	"EEG"	"Markers"	Content type. "EEG" enables automatic recognition by LabRecorder and BIDS tools.
channel_count	8	1	Must match the actual channel count. Marker streams always use 1 channel.
nominal_srate	1000.0	0	Sample rate in Hz. Use 0 (IRREGULAR_RATE) for event streams.
channel_format	cf_float32	cf_string	float32 for EEG data in μV ; string for human-readable event labels.
source_id	"inear_001"	"inear_001_markers"	Unique per-device identifier. Should remain stable across sessions.

8.2 XML Channel Metadata

The desc() XML tree embeds per-channel metadata that downstream tools use for labeling, unit conversion, and montage reconstruction:

Listing 3: Recommended LSL XML metadata for EEG stream

```
<channels>
  <channel>
    <label>EEG1</label>
    <unit>microvolts</unit>
    <type>EEG</type>
  </channel>
  <channel>
    <label>EEG2</label>
    <unit>microvolts</unit>
    <type>EEG</type>
  </channel>
  <!-- ... channels 3-8 ... -->
</channels>
<acquisition>
  <manufacturer>InEar Teensy</manufacturer>
  <model>Compact Protocol v1</model>
  <serial_number>DEVICE_SERIAL</serial_number>
  <scale_factor>0.02235</scale_factor>
</acquisition>
```

8.3 Important Metadata Fields

Table 13: LSL XML metadata fields that matter for interoperability

XML Path	Key	Example	Why It Matters
channels/channel	label	Fp1, Cz	Channel names for montage. 10–20 system names enable automatic montage in MNE-Python and EEGLAB.
channels/channel	unit	microvolts	Physical unit. Must be "microvolts" (not "uV") for consistent cross-tool behavior. ¹
channels/channel	type	EEG	Per-channel type. Enables mixed-type streams (e.g., channels 1–8 = EEG).
acquisition	manufacturer	InEar Teensy	Identifies hardware. Used for provenance tracking in BIDS datasets.
acquisition	model	Compact v1	Protocol / firmware version. Helps distinguish recordings made with different firmware.
acquisition	serial_number	INEAR_042	Hardware serial. Enables per-device calibration and multi-device setups.
acquisition	scale_factor	0.02235	μV per LSB. If the stream pushes raw ADC counts, tools need this to convert to microvolts.

8.4 Marker Stream Configuration

TTL events and experiment markers are sent on a separate irregular-rate string stream:

Table 14: Marker stream properties

Property	Value	Rationale
type	"Markers"	Standard LSL type for event streams
nominal_srate	0	Irregular rate — events arrive asynchronously
channel_format	cf_string	Human-readable labels
channel_count	1	Single event channel

Recommended marker string naming convention:

Table 15: Marker string formats

Event	Marker String
TTL input rising edge	"TTL_IN_1"
TTL input falling edge	"TTL_IN_0"
TTL output rising edge	"TTL_OUT_1"
TTL output falling edge	"TTL_OUT_0"
Experiment start	"EXPERIMENT_START"
Experiment stop	"EXPERIMENT_STOP"

8.5 Outlet Configuration

Table 16: LSL `StreamOutlet` parameters

Parameter	Value	Rationale
<code>chunk_size</code>	$\lfloor \text{srate}/20 \rfloor$ (= 50)	~50 ms chunks for latency / efficiency balance
<code>max_buffered</code>	360 (seconds)	6-minute buffer for brief network stalls

8.6 Metadata Checklist

Quick checklist for setting up a new EEG + Marker LSL outlet:

- ✓ `type` = "EEG" — not "eeg" or "Data".
- ✓ `nominal_srate` matches the actual hardware rate (1000.0).
- ✓ `channel_format` = `cf_float32` — push data in μV , not raw ADC counts.
- ✓ Every channel has a `label`, `unit` ("microvolts"), and `type` ("EEG").
- ✓ `acquisition` block includes `manufacturer`, `model`, `serial_number`.
- ✓ `source_id` is unique per physical device and stable across sessions.
- ✓ Marker stream uses `type` = "Markers", `srate` = 0, `cf_string`.
- ✓ `chunk_size` = `srate` / 20 for ~50 ms latency.